

Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol disclosed by two-colour sub-millisecond MINFLUX-based single-molecule tracking

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Article

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Concurrent diffusion of nicotinic acetylcholine receptors and fluorescent cholesterol
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Abstract

Nicotinic acetylcholine receptors (nAChRs) are ubiquitous neurotransmitter receptors predominantly located at the cell-surface of neurons and muscle cells. Their dynamics affect synaptogenesis at neurodevelopmental stages and the efficacy of synaptic transmission in the adult synapse. Here we exploit the enhanced capabilities of superresolution fluorescence MINFLUX microscopy to track for minute-long periods with nanometric precision and sub-millisecond time resolution the 2D translational dynamics of the bungarotoxin-labelled adult muscle-type nAChR in tandem with a fluorescent cholesterol analogue. To this end, we implemented a multiplexing procedure in continuous MINFLUX microscopy that enabled the simultaneous excitation of the two molecules using a single wavelength, followed by discrimination of their emissions via differential ratiometric recording. Single-molecule trajectories displayed a heterogeneous spectrum of diffusive behaviours (subdiffusive, Brownian and superdiffusive), with a predominance of the subdiffusive component, which became less pronounced upon cholesterol depletion. nAChRs spent most of their lifetime in confined areas of characteristic size ($\sim 0.005 \mu\text{m}^2$) lasting for ~ 100 ms. Further, MINFLUX captured regions where nAChR and fluorescent cholesterol moved jointly, both in confinement sojourns and along the free Brownian walks, which strongly indicated mutual interactions between the receptor macromolecule and the neutral lipid. To the best of our knowledge, this study constitutes the first series of experiments showing the diffusion dynamics of a transmembrane protein -a functionally important neurotransmitter receptor- together with a key membrane lipid in the native plasma membrane of a live cell at such high detail, thanks to the MINFLUX-based recordings.

Keywords: acetylcholine receptor; cholesterol; dynamics; diffusion; translational motion; superresolution microscopy; multiplexed MINFLUX; co-tracking.

Introduction

Superresolution microscopy methods are increasingly being successfully applied to unravel a wide range of biologically relevant topics (Sahl et al. 2017; Schermelleh et al. 2019). Neurobiology has benefitted from the inception of these approaches (Werner et al. 2021), enhancing the spatial and temporal resolution with which we can address the structure and dynamics of the neuron, the patterned compartmentalization of the axon, and the topography of the macromolecular constituents of the synapse in health (Hindley et al. 2023) and disease (Barrantes 2024a).

Single-particle tracking (SPT) or single-molecule tracking (SMT) techniques are among the biophysical methods of choice to elucidate the spatial extension and temporal duration of dynamic molecular processes involved in protein and cellular function at large. Because of the time window and constrained volume in which they occur, cell-surface phenomena like those involving neurotransmitter receptors are particularly challenging to study. The combination of single-molecule localization microscopy (SMLM) (Manzo and Garcia-Parajo 2015; Krapf 2015), DNA-PAINT microscopy (Niederauer et al. 2023; Deguchi et al. 2023), or the MINFLUX microscopy technique (Balzarotti et al. 2017; Carsten et al. 2022; Wolff et al. 2023; Deguchi et al. 2023; Eilers et al. 2018) with SMT methods are beginning to unravel the dynamics of synaptic components.

MINFLUX microscopy, one of the most recent superresolution microscopy methods, combines concepts from stimulated emission microscopy (STED) (Hell et al. 2015), a targeted approach, and a stochastic localization superresolution technique like SMLM (Huang et al. 2008; Heilemann et al. 2008). This powerful combination drastically reduces by a factor of ~100-fold the number of photons needed to localize individual molecules in comparison to camera-based imaging techniques (Balzarotti et al. 2017; Eilers et al. 2018). MINFLUX-based SMT is accomplished by employing a patterned doughnut-shaped illumination scheme that initially probes around the estimated emitter position, followed by a

computationally driven rapid refocusing of the patterned illumination, zooming in to the fluorescently labelled single molecule to “corral” and “lock” it to follow its trajectory.

Previous studies using camera-based fluorescence microscopy of α -bungarotoxin (BTX)-labelled nAChR have revealed its complex and heterogenous diffusion, sensitive to the cholesterol content of the plasma membrane (Almaraz et al. 2014; Mosqueira et al. 2018, 2020). Because of its inherently superior spatiotemporal resolution, MINFLUX microscopy should provide a deeper insight into the receptor dynamics. So far, however, MINFLUX has not been employed for the study of molecular dynamics in live-cell membranes. Further, and more importantly, it is only possible to image one fluorescent emitter at a time with this technique, thus precluding molecule co-tracking studies. As a remedy, the combination of FRET and MINFLUX microscopy as a means to explore short distances was suggested by Hell and coworkers (Eilers et al. 2018). Experimentally, a recent approach for observing two fluorescent emitters at a time was the combined use of pulsed MINFLUX microscopy (Masullo et al. 2021) with fluorescence energy transfer (FRET); in a proof-of-concept work, co-tracking of DNA origami labelled with two dyes differing in their fluorescence lifetimes was achieved (Cole et al. 2024). Here we overcome these issues by introducing a simple multiplexing variant in continuous excitation MINFLUX microscopy, using single wavelength excitation and a ratiometric differential emission detection. In this manner, we exploit the enhanced spatio-temporal resolution and minimal photon budget of MINFLUX to simultaneously follow with nanometric precision and sub-millisecond time resolution the joint single-molecule trajectories of a fluorescent cholesterol analogue and the nAChR protein, affording direct visualization of interactions between the neurotransmitter receptor and the neutral lipid at the cell surface of a live mammalian cell.

Results

Two-colour MINFLUX SMT of nicotinic receptor and cholesterol single-molecule trajectories with sub-millisecond time-resolution

CHO-K1/A5 cells are a clonal cell line robustly expressing adult muscle-type nicotinic acetylcholine receptor (nAChR) (Roccamo et al. 1999). A set of cells was stained with very low (500 pM) CF®640-labelled α -bungarotoxin (CF®640-BTX). After initial inspection of the cells in the confocal mode, multiple single-molecule tracks at a maximal rate of ~10 kHz were recorded (see Methods) from at least 10 cells per experimental condition within a usually 10 μm^2 large region-of-interest (ROI). This allowed us to accumulate in the order of several thousand single-molecule tracks within a few minutes (Figure 1a). The localization precision was calculated to be $\sigma = 7 \pm 1$ nm, and the time resolution was 492 ± 780 μs (see Methods). A second experimental dataset consisted of CHO-K1/A5 cells supplemented with 1 nM of a fluorescent cholesterol analogue, which was conjugated with the organic dye Abberior STAR Red via a polyethyleneglycol linker (fPEG-Chol). For the co-diffusion experiments we double-stained the CHO-K1/A5 cells with fluorescent α -bungarotoxin and fPEG-Chol. The spectral overlap of fluorescence emission between the two fluorophores Abberior STAR Red (labelling fPEG-Chol) and CF®640-BTX (labelling nAChR) was too high to distinguish them by colour. We therefore switched to an α -bungarotoxin labelled with a different fluorophore, CF®680R (CF®680R-BTX), which emits a more reddish colour, more easily distinguishable from fPEG-Chol. This choice enabled us to excite the two probes with the same laser line at a single wavelength and discriminate between the two probes by ratiometric measurement of their fluorescence emission wavelengths. To this end we established a detector channel ratio (DCR) criterion, as described in Suppl. Figure 1. We denoted fPEG-Chol trajectories as fPEG-Chol(+CF®680R-BTX) when we analysed the trajectories of the cholesterol probe in the presence of CF®680R-BTX and, reciprocally, CF®680R-BTX(+fPEG-Chol) when we followed CF®680R-BTX trajectories in the presence of fPEG-Chol (Figures 1b,c). Control experiments with CF®680R-BTX alone or with fPEG-Chol alone were also carried out to determine the DCR values of each individual probe and simply denoted CF®680R-BTX and fPEG-Chol, respectively.

Trajectory characterization and exclusion of immobile trajectories

Preliminary analysis of the data involved measuring trajectory and individual step durations (Suppl. Table 1), which however did not reveal further details of the diffusional behaviour of the probes. To be able to characterize the trajectories in more depth, the criterion of reference (Golan and Sherman 2017) was applied next to the determination of the proportion of immobile/mobile trajectories, as shown in Suppl. Figure 2. Using this criterion, about half of the trajectories of toxin-labelled nAChRs were classified as completely immobile. Notably, only ~20% completely immobile trajectories were observed in the case of fPEG-Chol tracks co-labelled with CF®680R-BTX. Since we wanted to disclose the actual (co-) diffusion modes of toxin-labelled nAChR and fluorescent cholesterol, and our previous analysis on general SMT tracks (including MINFLUX-based) highlighted biases for completely immobile events (Vogler et al. 2023), we focus here only on completely mobile trajectories and excluded immobile trajectories from subsequent analyses.

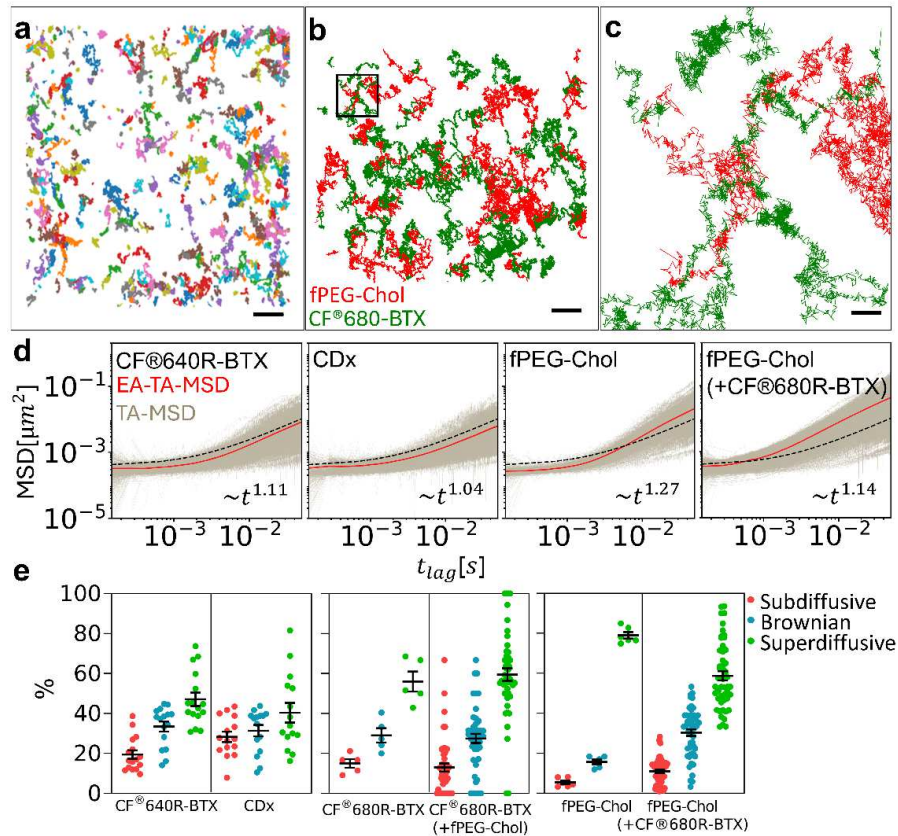


Figure 1. MINFLUX single-molecule tracking of individual bungarotoxin-labelled nicotinic acetylcholine receptors and fluorescent cholesterol in live cells, ensembled-averaged (EA), time-averaged (TA) mean squared displacement (EA-TA-MSD) and TA-MSD analyses, 2D-diffusion parameters and diffusive regimes derived from the anomalous coefficient β . a) Example of a 10 x 10 μm region of interest (ROI) image of the clonal cell line CHO-K1/A5 expressing adult muscle-type nAChRs labelled with CF[®]640-BTX only. Single-molecule tracks are randomly colour-coded as they appear on the screen of the MINFLUX microscope setup; after a few minutes, the ROI was covered with individual traces. Approximately 50% of the fluorescently labelled receptors appeared as nanometric-sized dots with no indication of motion. Subsequent analysis identified them as immobile nAChRs (Suppl. Fig. 2). Scale bar = 1 μm . b) Representative fPEG-Chol and CF[®]680-BTX-labelled nAChR trajectories in a co-tracking experiment. Scale bar = 500 nm. c) Zoom-in into marked region from (b). Scale bar = 100 nm. d) MSD analyses from different samples as marked (CDx: cholesterol depletion via methyl- β -cyclodextrin), TA-MSDs, EA-TA-MSD, and ideal Brownian diffusion (see Eq. 5 in Methods, where R, the correction factor for blurring was set to 1/6.2 (Balzarotti et al. 2017), and $\Delta = 10$ nm). Both X- and Y-axes are in log scale. e) Boxplot showing the percentage of trajectories classified as subdiffusive ($\beta < 0.9$), Brownian ($0.9 \leq \beta \leq 1.1$), and superdiffusive ($\beta > 1.1$) for CF[®]640R-BTX (n = 1,383 trajectories with an average of 1,105 steps, 16 ROIs), cholesterol-depleted CF[®]640R-BTX-labelled samples (CDx, n=853 and average 793 steps, 15 ROIs), CF[®]680R-BTX (n=187 and average of 925 steps, 5 ROIs), CF[®]680R-BTX(+fPEG-Chol) (n=1,373 and average of 2,034 steps, 47 ROIs), fPEG-Chol (n=1,012 and average of 1363 steps, 6 ROIs), and fPEG-Chol(+CF[®]680R-BTX) (n=5,229 and average of 1,496 steps, 52 ROIs). TA-MSD and EA-TA-MSD fittings extended up to 50 ms.

Quantitative analysis of nAChR diffusive behaviour

The ensemble-averaged, time-averaged mean squared displacement (EA-TA-MSD) is a time-dependent measure of the deviation of a particle position relative to a reference position. Here, EA-TA-MSD curves (Figure 1d) were furnished by fitting the individual curves with the analytical expressions (Eqs. 3-4) outlined in Methods, providing information on the ensemble diffusive behaviour of the nAChR macromolecules and the cholesterol analogue in a 2D Euclidean space -the membrane- over time.

MINFLUX allowed us to analyse a hitherto unexplored time window, i.e. the sub-millisecond time range (see Figure 1d). As a comparison, previous camera-based studies had a temporal resolution of ≈ 10 ms (Mosqueira et al. 2018, 2020). Table 1 lists the kinetic parameters derived from this analysis, i.e., the generalized diffusion coefficient (K_β), the apparent anomalous coefficient (β), the average duration (or residence time) of confinement sojourns, the average duration (or residence time) of non-confined portions, and the confined ratio r , which is defined as the quotient between the residence time of the trajectory in the confined state (i.e., the sum of the confined portions of a single trajectory) and the total duration of the trajectory. These parameters are for the ensemble population of trajectories under various experimental conditions, including cholesterol depletion (CDx), unlabelled cholesterol supplementation, and disruption of the submembrane actin cytoskeleton using CK-666 (inhibitor of the Arp2/3 complex mediating actin branching). The decrease in the value of K_β of the cholesterol-depleted (CDx) samples relative to the control CF[®]640R-BTX-labelled nAChRs ($p < 0.01$), indicates that extraction of membrane cholesterol slowed down nAChR diffusion, whereas in the presence of fPEG-Chol diffusion was 2-fold faster ($p < 0.01$). Analogously, fPEG-Chol diffused ~ 1.4 -fold faster in the presence of the toxin-labelled nAChR (fPEG-Chol(+CF[®]680R-BTX) samples; $p < 0.01$). Table 1 also shows the feeble effect of 50 nM exogenous cholesterol supplementation on toxin-labelled nAChR and fPEG-Chol diffusion. Only at 100 nM cholesterol addition did the diffusion coefficients increase, albeit modestly ($p < 0.01$). Table 1 also provides the kinetic parameters of the diffusion following perturbation of the actin cytoskeleton, which will be discussed in more detail in a later section.

Table 1*. Generalized diffusion coefficient (K_β), apparent anomalous coefficient (β) and confined ratio of the SMTs, effects of cholesterol depletion or enrichment mediated by methyl- β -cyclodextrin (CDx), and modification of trajectory parameters upon inhibition of the Arp2/3 complex by CK-666

Experimental condition	Mean $\pm$ S.E.M.				
	K_β ($\mu\text{m}^2\text{s}^{-\beta}$)	β	Average duration of confinement sojourn [ms]	Average duration of non-confined portion [ms]	#Confined ratio, r
CF [®] 640R-BTX-labelled nAChR	0.301 $\pm$ 0.023	1.09 $\pm$ 0.01	71 $\pm$ 8	12 $\pm$ 1	0.76 $\pm$ 0.02
Cholesterol analogue (fPEG-Chol) alone	1.357 $\pm$ 0.082	1.26 $\pm$ 0.01	25 $\pm$ 2	9 $\pm$ 1	0.63 $\pm$ 0.01
Far-red toxin (CF [®] 680R-BTX)-labelled nAChR alone	0.428 $\pm$ 0.017	1.16 $\pm$ 0.02	66 $\pm$ 8	20 $\pm$ 2	0.63 $\pm$ 0.01
fPEG-Chol in the presence of far-red (CF [®] 680R-BTX) toxin	1.710 $\pm$ 0.045	1.13 $\pm$ 0.01	8 $\pm$ 2	7 $\pm$ 1	0.49 $\pm$ 0.01
Far-red toxin (CF [®] 680R-BTX)-labelled nAChR in the presence of fPEG-Chol	0.926 $\pm$ 0.052	1.14 $\pm$ 0.02	39 $\pm$ 5	10 $\pm$ 1	0.67 $\pm$ 0.01
<i>Cholesterol-induced changes in kinetic parameters</i>					
CF [®] 640R-BTX-labelled nAChR (cholesterol depleted via CDx)	0.189 $\pm$ 0.016	1.06 $\pm$ 0.02	137 $\pm$ 24	15 $\pm$ 1	0.75 $\pm$ 0.02
CF [®] 640R-BTX-labelled nAChR + 50 nM cholesterol	0.517 $\pm$ 0.074	1.20 $\pm$ 0.05	55 $\pm$ 10	12 $\pm$ 2	0.72 $\pm$ 0.05
fPEG-Chol in the presence of far-red toxin (CF [®] 680R-BTX) + 50 nM cholesterol	2.009 $\pm$ 0.338	1.15 $\pm$ 0.02	11 $\pm$ 4	9 $\pm$ 1	0.45 $\pm$ 0.07
Far-red toxin (CF [®] 680R-BTX)-labelled nAChR+ fPEG-Chol + 50 nM cholesterol	2.127 $\pm$ 0.700	1.25 $\pm$ 0.07	27 $\pm$ 6	17 $\pm$ 4	0.53 $\pm$ 0.06
fPEG-Chol in the presence of far-red (CF [®] 680R-BTX) + 100 nM cholesterol	2.163 $\pm$ 0.211	1.13 $\pm$ 0.01	9 $\pm$ 1	7 $\pm$ 1	0.50 $\pm$ 0.02
Far-red toxin (CF [®] 680R-BTX) in the presence of fPEG-Chol + 100 nM cholesterol	1.451 $\pm$ 0.362	1.18 $\pm$ 0.03	40 $\pm$ 5	11 $\pm$ 1	0.71 $\pm$ 0.02
<i>Perturbation of the submembrane actin cytoskeleton network by CK-666</i>					
fPEG-Chol alone upon CK-666 treatment	1.751 $\pm$ 0.176	1.18 $\pm$ 0.01	16 $\pm$ 2	7 $\pm$ 1	0.58 $\pm$ 0.03
fPEG-Chol in the presence of far-red (CF [®] 680R-BTX) upon CK-666 treatment	1.931 $\pm$ 0.100	1.11 $\pm$ 0.01	9 $\pm$ 1	8 $\pm$ 1	0.47 $\pm$ 0.02
Far-red toxin (CF [®] 680R-BTX) upon CK-666 treatment	0.398 $\pm$ 0.064	1.13 $\pm$ 0.03	96 $\pm$ 34	16 $\pm$ 3	0.66 $\pm$ 0.03
Far-red toxin (CF [®] 680R-BTX) in the presence of fPEG-Chol upon CK-666 treatment	1.117 $\pm$ 0.182	1.15 $\pm$ 0.01	31 $\pm$ 2	17 $\pm$ 3	0.61 $\pm$ 0.01

*Exogenous unlabelled cholesterol depletion/supplementation data and CK-666 treatment are included in this table to facilitate comparisons and avoid repetition of control results. #Confined ratio r is defined as the quotient between the residence time of the trajectory in the confined state divided by the total duration of the trajectory.

The power (anomalous) exponent β obtained by fitting the individual time-averaged MSD (TA-MSD) curves was next used to classify trajectories according to their motional regime. Tracks with $\beta < 0.9$ were rated as subdiffusive, i.e. mobility was transiently confined or trapped, with $0.9 \leq \beta \leq 1.1$ as Brownian

i.e., free diffusion, and $\beta > 1.1$ as superdiffusive, i.e., deviating from free diffusion due to enhanced mobility due to e.g., active cellular transport processes. Both nAChR and the cholesterol analogue covered a wide spectrum of diffusive behaviours, as shown in Figure 1e. CF[®]640R-BTX-or CF[®]680R-BTX-labelled nAChRs trajectories exhibited a predominantly (~47-60%) superdiffusive component, ~33% Brownian, and ~20% subdiffusive ($p < 0.05$), with an increased subdiffusive behaviour (~30%) upon cholesterol depletion ($p < 0.05$). fPEG-Chol depicted a completely different, overwhelmingly (~80%; $p < 0.0001$) superdiffusive behaviour, which diminished in samples co-labelled with CF[®]680R-BTX (~60%; $p < 0.05$), concomitant with an increase in subdiffusive ($p < 0.05$) and Brownian ($p < 0.001$) trajectories. Additionally, the values of β and K_β corresponding to each of the three diffusion regimes analysed in this work are listed in Suppl. Table 2. The large fraction of superdiffusive trajectories resulted from the rather low threshold of β (> 1.1 ; several previous SMT analyses used much higher values (> 1.5) for this classification). This enabled us to highlight tendencies towards faster diffusion, e.g., due to larger fractions of more fluid (or less ordered) membrane regions during the track.

Dissection of trajectory properties in non-confined (free walks) and confined states

The dynamics within the free and confined states of the trajectories were next dissected separately (Figure 2a). The free, non-confined segments of CF[®]640R-BTX-labelled nAChR lasted 12 ± 1 ms, while the confined segments lasted significantly longer (71 ± 8 ms) (Figure 2b). The presence of fPEG-Chol in co-labelled samples significantly decreased the duration of both non-confined (20 ± 1 ms to 10 ± 1 ms, $p < 0.05$) and confined (66 ± 8 ms to 39 ± 5 ms, $p < 0.05$) states of CF[®]680R-BTX-labelled nAChRs. Cholesterol depletion (CDx) had the opposite effect, lengthening the duration of the confined portions to 137 ± 25 ms ($p < 0.01$). This was confirmed by experiments in which unlabelled cholesterol was added to the toxin-labelled samples, where the duration of the confined portions was reduced to a similar extent as that produced by fPEG-Chol (27 ± 6 ms at 50 nM, $p < 0.01$ Suppl. Figure 3).

To quantitatively assess the time spent by a trajectory in the confined state, a “confined ratio”, r (Table 1 and Figure 2b) was operationally defined as the quotient between the residence time of the trajectory in the confined state divided by the total duration of the trajectory. r was 0.76 ± 0.02 and 0.63 ± 0.03 for CF®640R-BTX and CF®680R-BTX, respectively, indicating that, on average, nAChRs spend most of their lifetime in confinement, as was the case with fPEG-Chol ($r = 0.63 \pm 0.02$) alone. The latter value decreased to 0.49 ± 0.02 in cells co-labelled with toxin (fPEG-Chol(+CF®680R-BTX)), $p < 0.01$).

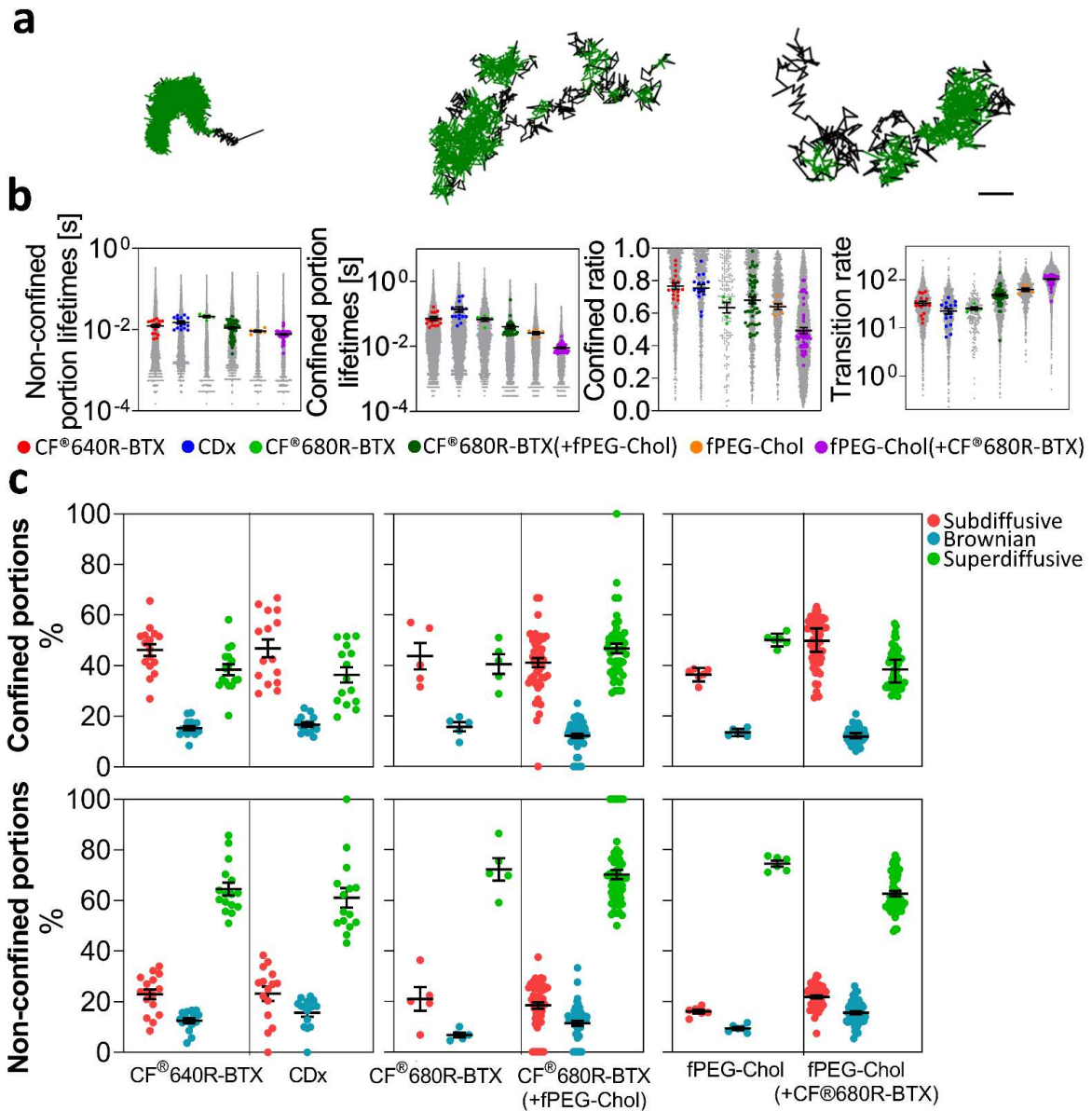


Figure 2. Confined and non-confined states of the cholesterol probe and toxin-labelled nAChR trajectories and their corresponding diffusive behaviour. a) Representative single tracks of CF[®]640R-BTX-labelled nAChR under control conditions showing parts classified as confinement sojourns (green) and as free, non-confined portions (black). Scale bar = 100 nm. b) Lifetimes of non-confined portions (left) and confinement sojourns (second left), confined ratio r (second right, defined as before $r = \text{confined portion lifetimes} / \text{total trajectory time}$), and transition rates between confined and non-confined states of the trajectories of CF[®]640R-BTX, CDx, CF[®]680R-BTX, CF[®]680R-BTX(+fPEG-Chol), fPEG-Chol, and fPEG-Chol(+CF[®]680R-BTX). Each grey dot represents a trajectory inside a ROI and the coloured dots represent the average of the parameter for a given ROI. c) Boxplots showing the mean and SEM of the percentage of trajectories classified as subdiffusive ($\beta < 0.9$), Brownian ($0.9 \leq \beta \leq 1.1$), and superdiffusive ($\beta > 1.1$) in confinement sojourns (upper panels) and non-confined portions (lower panels), respectively, upon TA-MSD fitting up to 25 ms for the different sample conditions as labelled.

Intuitively, if a molecule spends less time under confinement, one expects it to spend more time in the complementary non-confined part of the track. However, both non-confined and confined portions of CF[®]680R-BTX trajectories were shortened under fPEG-Chol co-labelling conditions (Figure 2). A possible explanation for this paradoxical observation is that the *transition rate* (i.e., the number of confined/non-confined state changes per unit time) (Figure 2b right) increased in the presence of cholesterol. Indeed, the transition rate of CF[®]680R-BTX increased from $25.49 \pm 1.93 \text{ s}^{-1}$ to $48.27 \pm 2.99 \text{ s}^{-1}$ in the presence of fPEG-Chol ($p < 0.01$). The transition rate of fPEG-Chol alone was $62.94 \pm 4.79 \text{ s}^{-1}$, augmenting to $103.70 \pm 2.68 \text{ s}^{-1}$ in the presence of CF[®]680R-BTX ($p < 0.001$). These results indicate that CF[®]680R-BTX changed state faster in the presence of fPEG-Chol and vice versa. Finally, supplementation with unlabelled cholesterol (50-100 nM) did not change the *total* time that the nAChR spent exploring each state, but rather changed the average residence time in confinement, which decreased from $0.84 \pm 0.09 \text{ s}$ to $0.34 \pm 0.09 \text{ s}$ upon cholesterol supplementation.

We next analysed the free moving and confined portions of the trajectories more closely by computing the TA-MSD of each portion (Michalet 2010), i.e. we determined values of the generalized diffusion

coefficient (K_β) and the apparent anomalous coefficient (β) for the non-confined and confined portions of the trajectories. The values are listed in Suppl. Table 3. The values of β again allowed us to rate the trajectory portions as either subdiffusive ($\beta < 0.9$), Brownian ($0.9 \leq \beta \leq 1$) or superdiffusive ($\beta > 1.1$), and determine the respective percentages, as done before for the whole tracks (compare Figures 1e and 2c). As shown in Figure 2c, the percentage of subdiffusive motion was higher in the confinement sojourns and, in the case of fPEG-Chol and CF[®]680R-BTX, the predominant behaviour in the non-confined state was in all cases superdiffusive. Both behaviours were to be expected, since confinement is a form of subdiffusion, while non-confinement is characterized by fast diffusion. The values of K_β followed similar trends: in the non-confined state of the nAChR, K_β was ~3 times faster in the presence of fPEG-Chol ($p < 0.05$). Cholesterol depletion or supplementation modified the diffusion properties in both confined and non-confined regions, as highlighted in Suppl. Table 3 and Suppl. Figure 3. Finally, Suppl. Figure 4 depicts additional metrics of the trajectories, such as the shape and size of a confinement area (in form of the total area, eccentricity and length of the major axis), and the number of observed steps of the trajectories within and between confinement sojourns. Further details on all the parameters are discussed next to the respective supplementary figures and tables.

Turning angle analysis

The high spatial and temporal resolution of the tracks, as given by MINFLUX microscopy, allowed us to use another metric to characterize the diffusional behaviour of nAChR and fluorescent cholesterol in their confined and non-confined states: the turning angles distended by the molecules along their SMTs. Specifically, we calculated the probability density function of turning angles between subsequent steps of a trajectory for different step lengths, as shown in Figure 5 for the unclassified and in Suppl. Figure 5 for the differently classified diffusion regimes (subdiffusive, Brownian, superdiffusive). The probability density function for Brownian motion is constant for angles between 90° to 180° (since there is no

turning angle predilection in this case), while there is an increase towards turning angles between 90° and 180° (i.e. anticorrelation) for confined or subdiffusive characteristic (due to molecules tending to move backwards), and turning angles between 0° and 90° (i.e. a positively correlated steps) for superdiffusive behaviours (as the molecule tends to move forward). Further, changes in the probability density function distribution for increased step lengths indicate the spatial extent of a confined, free-diffusive or superdiffusive portion of a track.

The turning angle analysis confirmed our previous analysis. The distribution for both CF®680R-BTX-labelled nAChRs and fPEG-Chol within confinement sojourns increased linearly as the turning angle approached 180°, i.e. both molecules tended to undergo anticorrelated steps. This confirms their prominently subdiffusive and rare Brownian-like or superdiffusive behaviour, as already depicted in Figure 2. Moreover, this preference for anticorrelated steps diminished as the step lag increased, suggesting the finite extent of confinement zones, thus hindering diffusion inside structures of characteristic size (Sadegh et al. 2017; Mosqueira et al. 2020). This finding was further supported by deep learning CONDOR analysis (Gentili and Volpe 2021), as shown in Suppl. Figure 6 and Suppl. Table 4; restricted mobility of both toxin-labelled nAChR *and* fPEG-Chol occurred predominantly within membrane compartments of specific sizes (0.006-0.008 μm^2).

In non-confined portions of the tracks, for short step lags, the motion was typical of obstructed diffusion (Weigel et al. 2012); for intermediate length steps, the turning angle distribution was uniform, characteristic of Brownian motion (Sadegh et al. 2017); and for long step lags, the molecules took highly correlated i.e. superdiffusive steps. This change in directionality of the turning angle distributions with variation of the step lag confirmed on one hand the heterogeneous characteristic of the diffusion even in the non-confined portions of the SMTs, and on the other hand the augmented superdiffusive motion in the non-confined state, as shown in Figure 3.

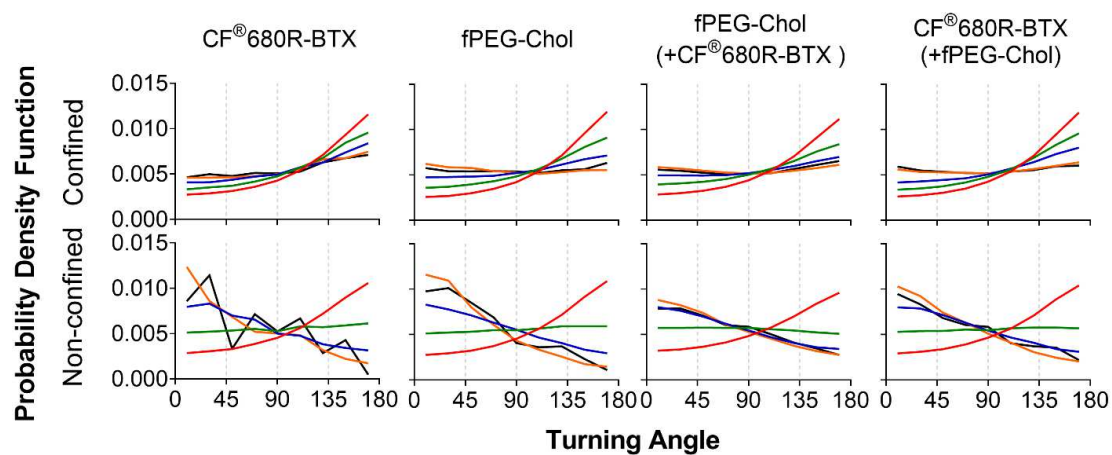


Figure 3. Breakdown of turning angle analysis of nAChR and cholesterol analogue trajectories discriminated into confined and non-confined, free-walking states: probability function (PDF) of the turning angles (0° to 180°) of the trajectories for the different samples and the two classifications as labelled. The differently coloured probability density functions correspond to 1 (*red*), 4 (*green*), 8 (*blue*), 25 (*orange*), 50 (*black*) step lags.

Temporal overlap between nAChR and cholesterol trajectories

In a next step we analysed simultaneously recorded trajectories of nAChR (labelled with CF®680R-BTX) and fPEG-Chol (labelled with Abberior STAR Red) probes with high detail, especially to disclose any co-diffusion events. As highlighted before, this choice of labels (CF®680R and Abberior STAR Red) enabled us to excite the two probes with the same laser line and discriminate between the two probes by ratiometric measurement of their fluorescence intensities in two spectrally separated detection channels. Specifically, we calculated the detector channel ratio (DCR) by division of the fluorescence intensities registered in the more reddish and blueish detection channels, respectively (Methods and Suppl. Figure 1). Control measurements on the two probes individually revealed clearly separated high and low DCR values for fPEG-Chol and CF®680R-BTX, respectively (Suppl. Figure 1). This is further depicted in the left two panels of Figure 4a, which highlight representative tracks and the temporal evolution of DCR and total intensity values (summed over both detection channels and given as the effective counts at offset (ECO). The latter correspond to the photon counts collected at the outer

points of the MINFLUX scanning pattern, corrected for background contributions for individually recorded fPEG-Chol and CF[®]680R-BTX. DCR values were above 0.55 in the case of CF[®]680R-BTX and below 0.40 for fPEG-Chol, and ECO/intensity values at very low levels. However, simultaneous recordings of both probes revealed a significant fraction of trajectories with DCR values between 0.4-0.55 and at the same time increased ECO/intensity values, disclosing sections of simultaneous co-tracking detection and thus co-diffusion of the two probes (Figure 4a). This is also revealed in the frequency histograms of the DCR values, revealing a shift of the DCR values towards the intermediate regime between 0.4-0.55 (Figure 4b). Figure 4b also clearly highlights that both modulation of the actin cytoskeleton by CK-666 and addition of the unlabelled cholesterol affected the co-tracking of the nAChR and fluorescent cholesterol.

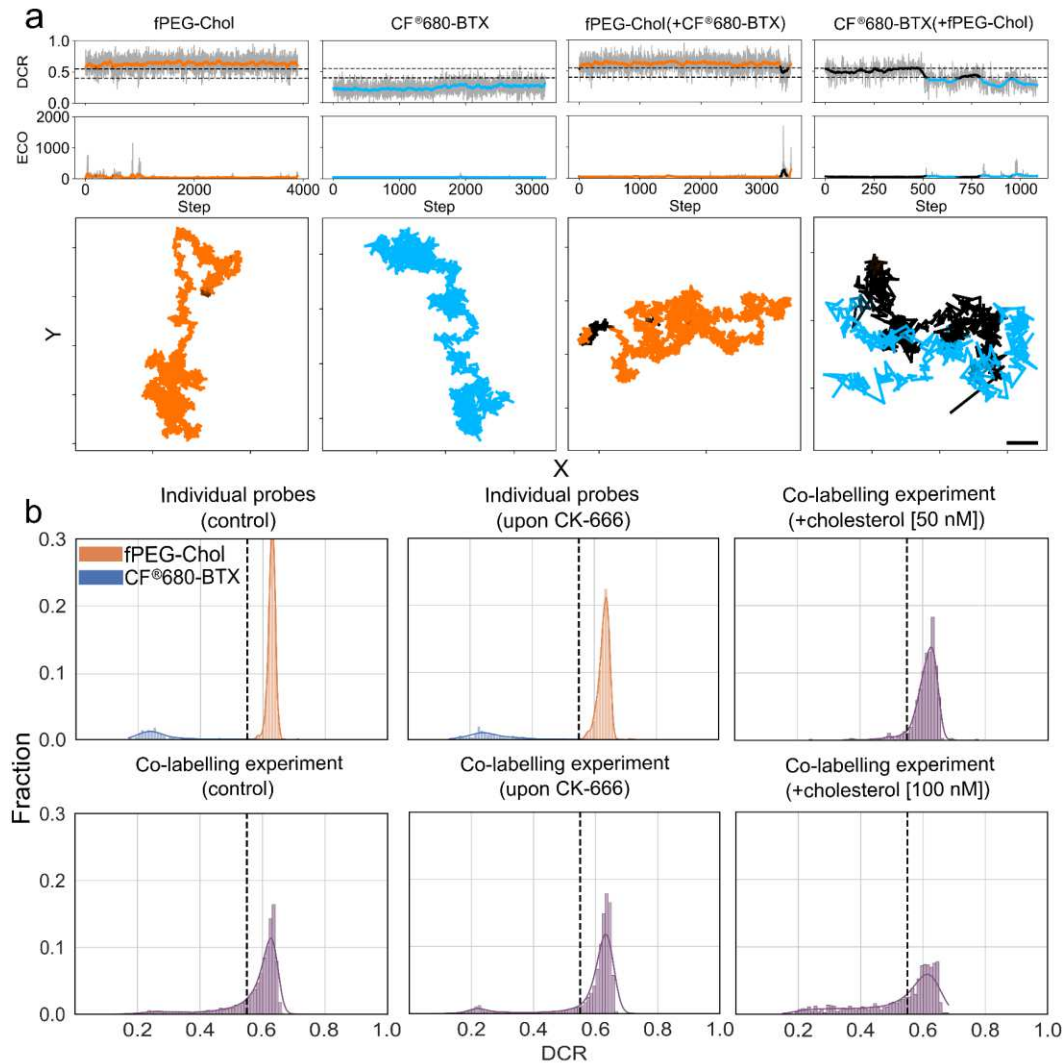


Figure 4. Co-diffusion analysis of fPEG-Chol and CF®680R-BTX-labelled nAChR using detector channel ratio (DCR) discrimination in combination with intensity changes. a) Representative data of DCR and intensity values (as effective counts at offset (ECO)) over time steps (upper two panels) and corresponding tracks in the X-Y plane with colours indicating classification as fPEG-Chol (orange), CF®680R-BTX (blue) and co-diffusion (black) for cells with fPEG-Chol alone, CF®680R-BTX nAChR alone, and the 2-colour co-labelling experiments with both fPEG-Chol and CF®680R-BTX). Co-diffusion becomes apparent in the increased ECO values and simultaneous change in DCR values converging in a corridor between 0.40 and 0.55 (marked by the two horizontal dashed lines Scale bar = 100 nm. b) Histograms of DCR values for the tracking data of fPEG-Chol (orange) and CF®680R-BTX (blue) only, and of fPEG-Chol and CF®680R-BTX co-labelled cells (purple) with two-species Gaussian fits for experiments with only one probe without (upper left, control) and with (upper middle) actin modulation by CK-666, and with

doubly-labelled experiments without any treatment (lower left, control), with actin modulation by CK-666 (lower middle) and with cholesterol supplementation (50 nM: upper right, 100 nM: lower right). The histograms built from the data acquired with both probes simultaneously show only the mobile fraction.

Quantitative assessment of spatial overlap between toxin-labelled nAChR and fPEG-Chol trajectories

Through visual inspection, we observed that trajectories of fPEG-Chol spatially intersected with those of CF[®]680R-BTX in confinement and non-confined zones, in several cases more than once along the same trajectory (Figure 5a, *left*). Nearly 70% of the ROIs were found to have a statistically significant overlap between confinement portions of nAChR and cholesterol trajectories ($p < 0.01$). To quantify this effect, an overlap coefficient (C) was calculated (see Methods). As shown in Figure 5b, in the confined region of the trajectories C was 0.27 ± 0.02 , indicating that, on average, about a quarter of the CF[®]680R-BTX confinement sojourns overlapped with those of fPEG-Chol. Addition of 100 nM unlabelled cholesterol drastically reduced the overlap by more than two-thirds ($C = 0.08 \pm 0.01$, $p < 0.001$). The overlap of non-confined portions was also found to be statistically different from chance: 5% of the ROIs were found to overlap ($p < 0.01$) along the free walk portions, with a C value of 0.40 ± 0.03 , indicating that the degree of overlap in non-confined portions was higher than in confined portions. Moreover, upon addition of 100 nM cholesterol, the non-confined overlap coefficient C was more drastically reduced to 0.13 ± 0.01 ($p < 0.001$), highlighting the important role of cholesterol in the co-diffusion; in this case the unlabelled cholesterol probably displaced the labelled fPEG-Chol, resulting in less detectable co-diffusion events.

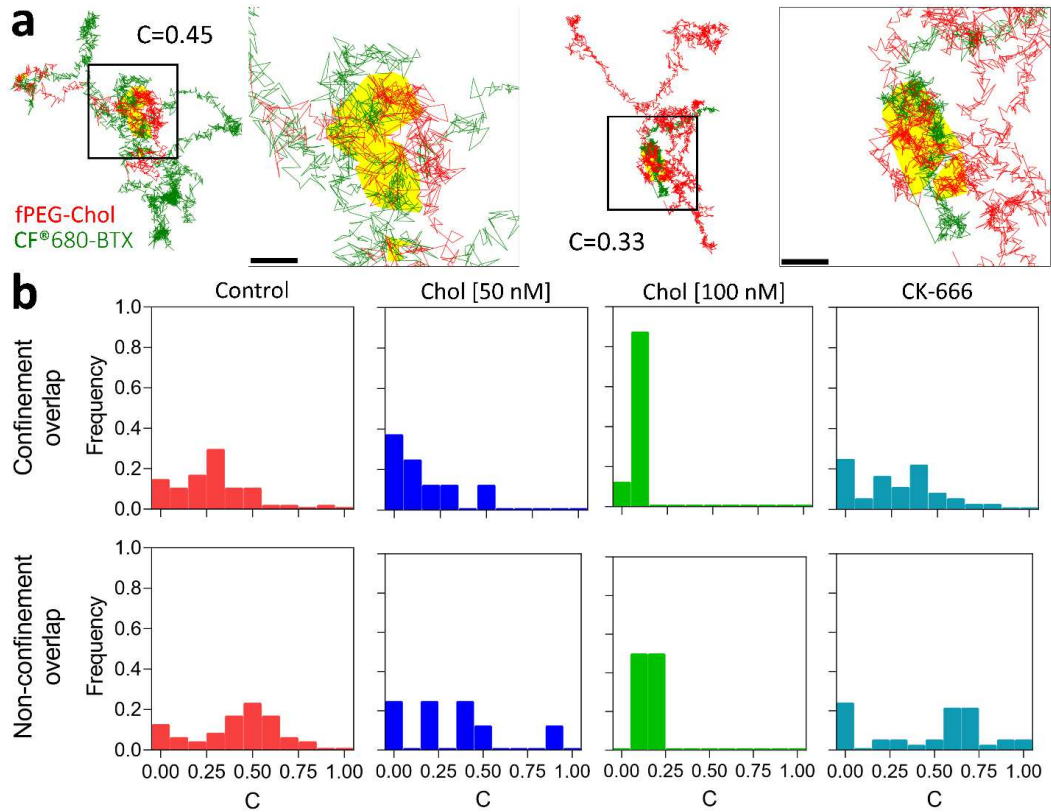


Figure 5. Spatial overlap of BTX-labelled nAChR and fluorescent cholesterol tracks in confinement sojourns and Brownian sections of the tracks, and effects of cholesterol addition and CK-666 treatment.

a) Two representative examples of trajectories of the two-colour labelled experiment with the track of the CF[®]680R-BTX-labelled nAChR and of Abberior STAR Red labelled fPEG-Chol: large overviews (left and middle right panels) and respective zoom-ins to the indicated black boxes (middle left and right). The confinement areas classified for CF[®]680R-BTX and of fPEG-Chol are highlighted in yellow. Their overlap coefficient C is indicated as a number in each case. Scale bar = 50 nm. b)

Histograms of overlap coefficient C (bin width = 0.1) for CF[®]680R-BTX and fPEG-Chol without treatment (left) following 50 nM (middle left) and 100 nM (middle right) cholesterol addition, and upon CK-666 treatment (right) in confinement (upper panels) and non-confinement (lower panels) areas.

Perturbation of the actin subcortical meshwork

To investigate the influence of the (cortical) actin cytoskeleton on the (co-)diffusion dynamics of nAChR and cholesterol, we incubated the samples of CHO-K1/A5 cells for 15 min at 4°C with CK-666, a compound that inhibits Arp2/3 complex formation (Pollard 2007; Hetrick et al. 2013) and hence the

branching and growth of new actin filaments, thus perturbing the stability of the submembrane (cortical) actin meshwork. The cortical actin cytoskeleton, i.e. the actin filaments underlying the plasma membrane, usually builds up a checkered-like network, “fencing in” compartments, where diffusion over an actin border from one compartment to the next is restricted, resulting in a compartmentalized or hop-like diffusion (e.g., (Fujiwara et al. 2016)). Previous experiments have also highlighted an effect of the actin cytoskeleton on nAChR organization (Wenz et al. 2010). The effects of CK-666 treatment on the kinetic diffusion parameters of CF®680R-BTX nAChR and fPEG-Chol are included in Table 1 to facilitate comparison with the other conditions. No changes were apparent in the anomalous diffusion coefficient β for any of the experimental conditions tested except for fPEG-Chol: upon CK-666 treatment, the fluorescent cholesterol analogue was less superdiffusive ($p < 0.05$). Only subtle changes were observed in the apparent diffusion constant K_β ; the fluorescent cholesterol probe alone was 1.1-times faster, and in the presence of CF®680R-BTX, 1.3-times faster, upon CK-666 treatment. The confined ratio r was not affected by CK-666 treatment, except for CF®680R-BTX in the presence of fPEG-Chol, in which case r decreased moderately ($p < 0.01$). The durations of the confined segments of the tracks were also affected: fPEG-Chol trajectories increased, and those of CF®680R-BTX decreased ($p < 0.01$) upon CK-666 treatment (Table 1). Interestingly, the drug did not affect the degree of overlap of fPEG-Chol and toxin-labelled nAChR in the confinement sojourns ($C = 0.27 \pm 0.01$) but did increase the degree of overlap in non-confined portions (0.46 ± 0.05 ; $p < 0.01$).

In order to investigate whether Arp2/3-mediated perturbation of the submembrane actin meshwork induced the so-called hop diffusion, we next undertook an analysis of the trajectories following the approaches of Hell and coworkers (Rickert et al. 2024) and Eggeling and coworkers (Reina et al. 2022). Here we classified the motion of molecules in three models: freely diffusing, confined, and hopping between compartments (Figure 6). For the hopping diffusion we introduced specific kinetic diffusion parameters of the average actin compartment diameter L , the diffusion coefficients D_μ and D_M

characterizing the short-term diffusion within the actin compartments (intra-compartmental) and long-term diffusion across compartments (inter-compartmental), respectively, and the average scale length L_{hop} of spatial separation between D_{μ} and D_M (Table 2). For both CF®680R-BTX and fPEG-Chol, only a small portion of the trajectories revealed hop-like diffusion (~15%); diffusion was rather dominated by free (39% and 30% for CF®680R-BTX and fPEG-Chol, respectively) and confined (39% and 30% for CF®680R-BTX and fPEG-Chol, respectively) diffusion. CK-666 treatment led to hardly any change in these numbers since the actin network was still intact but with less tight branching, i.e. larger compartment sizes, highlighted by the increase in values of L and L_{hop} from around 130-160 nm to 170-250 nm. Most significantly, CK-666 treatment resulted in overall enlargement of the diffusion coefficients: i) the diffusion coefficient D_{free} of the freely diffusing fraction increased by a factor of 2.8 for CF®680R-BTX and 1.6 for and fPEG-Chol; ii) mobility was also increased for the confined trajectories (a factor of 2.45 for CF®680R-BTX and 1.3 for fPEG-Chol); and iii) for hopping diffusion, both the intra- and inter-compartmental coefficients D_{μ} and D_M increased by a factor of 2-2.5. Despite this increase in D_{μ} and D_M , the compartmentalization strength $S = D_{\mu}/D_M$, which is inversely proportional to the hopping probability (i.e. the probability of crossing an actin-compartment boundary) (Reina et al. 2022), remained almost constant (3.07-3.46 for CF®680R-BTX and 2.43-3.45 for fPEG-Chol). This indicates that although the velocity of the molecules increased, they remained spatially constrained upon disturbing the cytoskeletal fences with CK-666, as highlighted above when discussing the negligible changes in fractions of diffusion modes and values of L and L_{hop} .

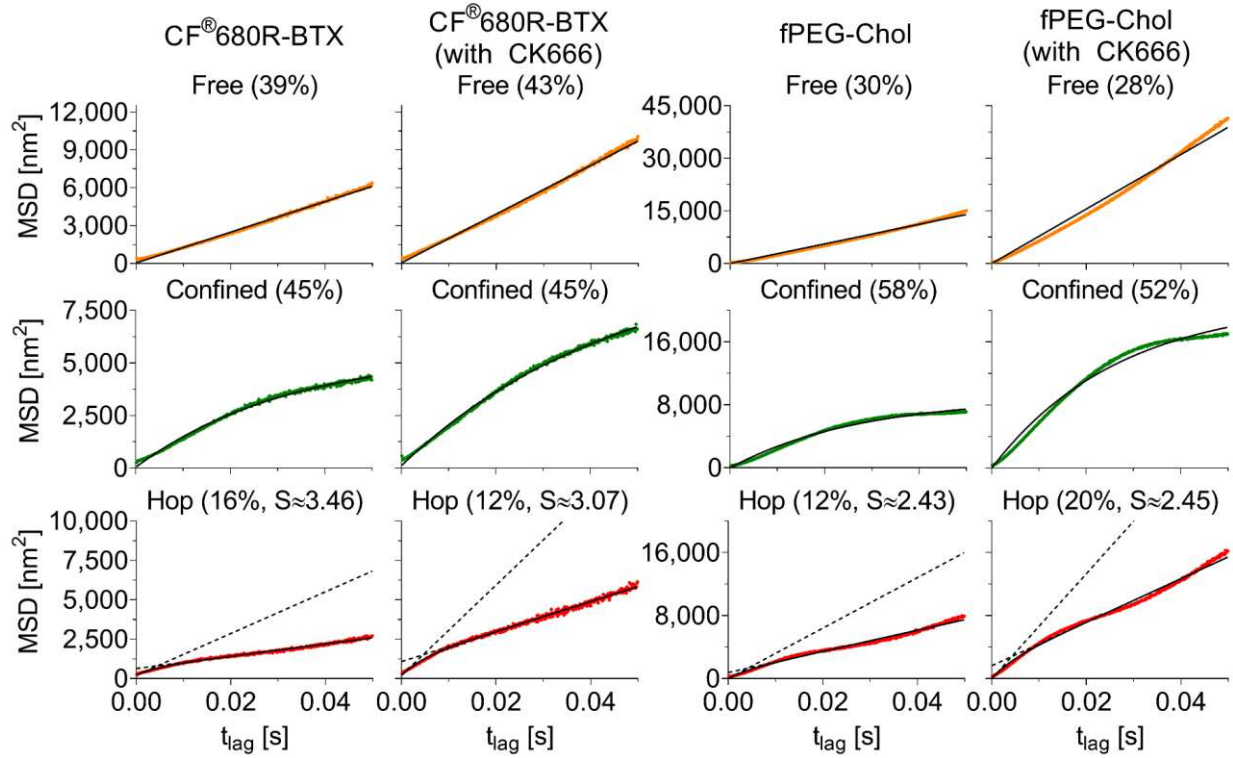


Figure 6. Analysis of diffusion data upon actin modulation. Ensemble-averaged, time averaged mean square displacement (EA-TA-MSD) curves determined for all trajectories (Brownian and subdiffusive) of CF®680R-BTX and fPEG-Chol with and without perturbation of the subcortical actin meshwork by CK-666, as labelled. The motional states were classified into three categories: free (orange, upper panels), confined (green, middle panels), and hop-like (red, lower panels) diffusion according to (Reina et al. 2022) and (Rickert et al. 2024). S denotes the confinement strength for the hopping diffusion, defined as the D_{μ}/D_M ratio, and the black lines in the panels of the upper two rows give the fits. The dotted lines in the panels of the lower row correspond to the expected MSD for pure diffusion with D_{μ} and D_M .

Table 2. SMT diffusion parameters of nAChR and fluorescent cholesterol trajectories corresponding to different diffusion models upon actin cytoskeleton perturbation with CK-666

Condition	$D_{free} [nm^2 s^{-1}]$	$D_{\mu} [nm^2 s^{-1}]$ in confinement	$D_M [nm^2 s^{-1}]$ between compartments	$D_{\mu} [nm^2 s^{-1}]$ within compartments	$L[nm]$	$L_{hop}[nm]$
fPEG-Chol	69,980	81,339	33,612	81,748	163.20	62.46
fPEG-Chol upon CK-666 treatment	194,595	196,997	69,116	169,990	251.80	90.93
Far-red toxin (CF®680R-BTX)	30,393	42,181	9,770	33,842	128.26	42.05
Far-red toxin (CF®680R-BTX) upon CK-666 treatment	48,267	54,158	23,813	73,193	172.88	62.34

D_{free} : average diffusion coefficient in the trajectories classified as free diffusing. D_{μ} in confinement: diffusion coefficient within confined sojourns of the trajectories. D_{μ} : in between compartments, i.e., short-time (intra-compartmental) diffusion coefficient for the trajectories classified as hopping diffusion. D_M : long-time inter-compartmental diffusion coefficient for the trajectories classified as hopping diffusion. L : apparent confinement domain size or, in this case, the average actin compartment size explored in the trajectories classified as hopping diffusion. L_{hop} : average scale length of separation between D_{μ} and D_M in the trajectories classified as hopping diffusion.

Discussion

We have applied MINFLUX nanoscopy to simultaneously follow the lateral motion of individual fluorescent-tagged cholesterol and neurotransmitter receptor protein molecules at a sampling rate of up to 10 kHz, i.e., at $\approx 600 \mu\text{s}$ per individual step, and with a localization precision $\sigma = 7 \pm 1 \text{ nm}$ (Table 3, Methods). This accomplishment required the development of a novel detection scheme that enabled the co-tracking of both the Abberior STAR Red labelled cholesterol probe (fPEG-Chol) and the nicotinic acetylcholine receptor (nAChR labelled with fluorescent BTX). The fluorescence emission of both probes was thereby excited simultaneously at the same wavelength and ratiometrically discriminated by emission wavelength, overcoming a bottleneck of conventional MINFLUX studies. Single-molecule trajectories of a fluorescent phospholipid, DPPE-ATTO 647N, in an artificial lipid bilayer (Schmidt et al. 2021) and the regular stepping motion of the kinesin protein motor on microtubules (Deguchi et al. 2023; Wirth et al. 2023; Wolff et al. 2023) have been documented with MINFLUX or MINSTED (Scheiderer et al. 2023), but to our knowledge sub-millisecond co-tracking localization studies of a membrane protein macromolecule together with a fluorescent lipid in the complex membrane environment of a living cell have not been reported to date.

Diffusion properties of the BTX-labelled nAChRs and the fluorescent cholesterol probe

Exploiting the superior time-resolution capability of the MINFLUX technique, we were able to dissect the diffusional properties of individual trajectories into their non-confined and confined portions along the tracks. nAChRs labelled with either CF[®]640-BTX or CF[®]680-BTX displayed a heterogeneous diffusion, ranging from subdiffusive (i.e. confined) over unhindered (i.e. random Brownian) to comparably very fast (i.e. superdiffusive) motion, as reported before in camera-based studies (Mosqueira et al. 2018). Subdiffusive motion became more pronounced upon cholesterol depletion, yet with a high proportion of superdiffusive components (Figure 1 and Table 1). This provided the first piece of evidence on the influence of the neutral lipid on the nAChR motility in this hitherto unexplored temporal window. The simplest explanation for the translational motion heterogeneity of the nAChR is the reported coexistence of dispersed individual macromolecules with nanoclusters of variable sizes. It has been reported that crowding of nAChRs in nanoclusters transiently hinders their motion, by a picket-like mechanism that could be accounted for in terms of a 2-state model switching between Brownian motion and obstructed diffusion (Mosqueira et al. 2018, 2020). This hindered motion applies both to individual molecules within the cluster as well as the assembly as a whole.

The fluorescent cholesterol analogue fPEG-Chol employed in this study was chosen because i) its polyethylene glycol chain maintains the fluorophore sufficiently far from the sterol moiety (Honigsmann et al. 2013), making PEG-labelled lipids, and fPEG-Chol in particular, among the least biophysically and physiochemically membrane perturbing probes (Mobarak et al. 2018; Sezgin et al. 2016); this ensures functional preservation of indirect, membrane-mediated cholesterol effects on the membrane and direct cholesterol-nAChR interactions (Barrantes 2024b); ii) it exhibits advantageous spectroscopic properties (with respect to absorption and emission wavelengths and photostability) for MINFLUX microscopy, and in particular for the 2-colour double-labelling scheme introduced here, and iii) it is known to reside almost exclusively at the cell-surface membrane, presumably at its outer, exoplasmic

leaflet (Pinkwart et al. 2019; Honigmann et al. 2013), with only a small fraction internalized by cells in the course of hours (Sezgin et al. 2016), well beyond the time window of our experiments.

As expected, the apparent diffusion of fPEG-Chol (alone or in the presence of the far-red CF[®]680R-BTX) was faster than that of the toxin-labelled nAChR protein (Table 1); the fluorescent cholesterol exhibited Brownian and superdiffusive and rarely subdiffusive motions (Figure 1 and Suppl. Table 2). MINFLUX further revealed that the motion of the nAChR, as assessed by its apparent anomalous diffusion constant K_β , was influenced by the presence of the fluorescent cholesterol probe (Table 1). While cholesterol depletion via methyl- β -cyclodextrin markedly slowed down nAChR diffusion, addition of 100 nM cholesterol only slightly accelerated diffusion (Table 1), indicating that the membrane cholesterol concentration is close to saturating concentrations in CHO-K1/A5 cells, as previously reported (Borroni et al. 2007). Taken together, this series of experimental results strongly indicates that the 2D translational motions of cholesterol and nAChR are correlated.

Cholesterol affects both the confined and non-confined portions of the nAChR SMTs. MINFLUX further afforded the spatiotemporal discrimination of the trajectories into their free diffusing segments and confinement sojourns, thus enabling calculation of the apparent diffusion coefficient in the two motional regimes. Whereas previous studies in cells and giant unilamellar vesicles yielded overall diffusion coefficient values of fPEG-Chol analogues close to $\sim 1 \mu\text{m}^2 \text{s}^{-1}$ (Pinkwart et al. 2019; Sevcsik et al. 2015), here we could discriminate between a slow component in the confined sojourns ($1.10 \pm 0.01 \mu\text{m}^2 \text{s}^{-1}$) and the superdiffusive, 6-fold faster ($6.45 \pm 0.24 \mu\text{m}^2 \text{s}^{-1}$) component in the free walks (Suppl. Table 3). The same applies to the dissection of the diffusion coefficients of the BTX-labelled nAChRs, differing by a factor of 5 between free diffusing portions and confinement sojourns (Suppl. Table 3).

The duration of the non-confined portions of the BTX-labelled nAChR SMTs under control and cholesterol depletion conditions were in the order of ~ 15 ms (Figure 2). These free-walking, random motion periods were statistically shorter in the presence of fPEG-Chol (10 ± 1 ms, $p < 0.01$), i.e., falling beyond the range of those of the cholesterol probe alone (9 ± 1 ms) or fPEG-Chol in the presence of CF[®]680R-BTX (7 ± 1 ms). These experimental results further reinforce the notion of mutual dynamic interactions between cholesterol and the receptor protein. The effect of the neutral lipid was also manifested on the confined sojourns: cholesterol depletion increased by 2-fold the lifetime of the confined periods from 71 ± 8 ms (control) to 137 ± 24 ms (CDx) ($p < 0.01$). fPEG-Chol confinement sojourns lasted one order of magnitude less (25 ± 1 ms) ($p < 0.0001$, Figure 2), with no changes in K_β in the presence of the toxin. The same applies to K_β in the non-confined portions of fPEG-Chol in the presence of the nAChR protein. However, confined portions became more subdiffusive (Figure 2) when fPEG-Chol was sampled in the presence of CF[®]680R-BTX ($p < 0.001$). Exogenous cholesterol supplementation only reduced the duration of the confined sojourns of CF[®]680R-BTX ($p < 0.01$). Cholesterol also affected the co-tracking of the two molecules (Figure 4). Undoubtedly, the main effect of cholesterol on nAChR diffusion is exerted on the confined portion of the nAChR trajectories. All in all, these kinetic parameters strongly suggest that the motion of nAChRs and cholesterol influence each other's diffusion behaviour.

Trajectory directional (turning angle) and machine learning CONDOR analyses

nAChRs change the direction of their trajectories by step-by-step angular displacement. If the steps display no preference for moving forward or backward, as in Brownian motion, there is no turning angle predilection. If a molecule favourably adopts anticorrelated steps (i.e., the molecule tends to move backwards), as in confined diffusion, turning angles between 90° and 180° are characteristically observed, whereas when steps are positively correlated (i.e., the molecule tends to move forward), as in

very fast i.e. superdiffusive motions, they vary between 0° and 90° . Here we measured step lags instead of time lags due to the irregular time intervals obtained with MINFLUX recordings. We were not only able to validate previous camera-based STORM studies (Mosqueira et al. 2018, 2020) but could also reveal that cholesterol accompanied the nAChR with a similar turning angle dependence. As expected, subdiffusive trajectories exhibited a strong preference for anti-correlated steps, a tendency that diminished as β increased (Suppl. Figure 5).

When turning angle analysis was dissected into confined and non-confined portions, anticorrelated steps were found in confinement sojourns, whereas no preference was observed in non-confined portions, which -as Figure 3 points out- exhibited superdiffusive, Brownian, and subdiffusive behaviour. The results of turning angle analysis showed that the model that best describes the heterogeneous behaviour of the nAChR is the two-state obstructed diffusion model (Mosqueira et al. 2020). Moreover, the strong step lag dependence of the probability density function suggested that structures of characteristic size were responsible for the obstructed diffusion (Sadegh et al. 2017; Mosqueira et al. 2020), a feature disclosed with better precision using the MINFLUX measurement (Suppl. Figure 4) and further confirmed by the deep learning CONDOR analysis (Suppl. Figure 6 and Suppl. Table 5). Consequently, the fluorescent cholesterol analogue followed the same motional conduct as the nAChR.

Overlap of nAChR and cholesterol single-molecule trajectories

One of the main findings of this work is the demonstration that fluorescent labelled nAChR and cholesterol molecules can be co-tracked for distinct periods along their trajectories (Figure 4). Using the overlap trajectory analysis outlined in Methods, it was further possible to quantitatively estimate the spatial overlap of these joint motional periods, i.e. fPEG-Chol and CF[®]680R-BTX overlapped 25% and ~40% in confinement sojourns and non-confined portions, respectively (Figure 5). Probabilistically, this represents a significantly high percentage ($p < 0.01$), considering the low concentration of CF[®]680R-BTX

labelled nAChRs (0.5 nM) and fPEG-Chol (1 nM) employed, the dimensions of the 2D host -the plasma membrane-, the size and diffusion coefficients of the incumbent molecules, and the length of their trajectories. Upon addition of exogenous unlabelled cholesterol, the overlap was reduced to ~8% with 100 nM cholesterol in confined regions, probably reflecting a law of mass action-type displacement of the fluorescent cholesterol by the 100-fold higher concentration of unlabelled sterol. Similar effects were observed on the non-confined portions of the trajectories.

The submembrane actin meshwork constitutes an additional hindrance to translational diffusion of nAChR and cholesterol

The actin compartment is the submembrane meshwork or actin-based membrane skeleton fence ("MSK"), and thus very relevant to cell-surface molecules like the neurotransmitter receptor and its interacting cholesterol molecule. The notion of such a specialized actin supramolecular structure was introduced by Sheetz (Sheetz 1983), who described the structure as "corrals" that restricted the lateral diffusion of membrane proteins, an idea subsequently examined in detail by Saxton and Jacobson (Saxton 1995; Saxton and Jacobson 1997). Computer-enhanced high-speed SPT enabled the Kusumi group (Fujiwara et al. 2002) to measure these corrals and extend the hypothesis to membrane lipids, leading in turn to the concept of time-dependent "hop diffusion": short-term confined diffusion within a compartment and infrequent, long-term hop diffusion between compartments of characteristic size (Ritchie et al. 2005), later supported by combined STED microscopy and fluorescence correlation spectroscopy (STED-FCS) data (Andrade et al. 2015)) and other SPT experiments based on e.g., interferometric SCATtering or fluorescence microscopy (Sadegh et al. 2017). In a previous STED superresolution imaging study, the cytoskeletal-disrupting drugs cytochalasin D and jasplakinolide were found to produce a statistically significant increase in the size of nAChR nanoclusters at the plasmalemma of CHO-K1/A5 cells, suggesting that the submembrane actin compartment affected the receptor distribution and local density at the cell surface (Wenz et al. 2010).

CK-666 is a small molecule that stabilizes the inactive state of Arp2/3, a protein complex involved in the nucleation of branched actin filaments that form the actin MSK corrals (Pollard 2007; Hetrick et al. 2013). Treatment with CK-666 resulted in a modest increase in diffusion of both nAChR and fluorescent cholesterol, implicating the contribution of cytoskeletal fences or corrals (Fujiwara et al. 2016) in addition to the pickets of clustered nAChR molecules that “self-restrict” the protein’s autologous diffusion at the plasmalemma (Mosqueira et al. 2018, 2020). SPT measurements in different mammalian cells (Murase et al. 2004; Andrade 2015) (Reina et al. 2022; Chai et al. 2022) showed that the actin MSK corrals have mesh sizes in the 40-100 nm range, in agreement with the current MINFLUX measurements showing apparent dimensions in the order of 50 nm. The 1.5-fold acceleration of the fast cholesterol diffusion upon CK-666-mediated actin disruption (Table 1) agrees with literature values: fluorescent cholesterol is only modestly influenced by actin corrals (Hiramoto-Yamaki et al. 2014; Sezgin et al. 2016). While the diffusion regime remained unaltered, the percentage of the free diffusing component increased (Figure 5), as did the anomalous diffusion constant K_β (Table 1) and the distribution of joint co-tracking of nAChR and fluorescent cholesterol (Figure 4). When analysed in terms of the hop-diffusion model using recently introduced criteria for MINFLUX studies (Rickert et al. 2024), both the short-term intra-compartmental (D_μ) and long-term inter-compartmental (D_M) diffusion coefficients increased upon CK-666 treatment (Figure 6 and Table 2), suggesting disruption of corral fences. Interestingly, despite the increase in the diffusion coefficients of toxin-labelled nAChR and fPEG-Chol, the degree of overlap between the two molecules did not change, another manifestation of the interaction between the two within confinement sojourns and while crossing compartments. Altogether the CK-666 experiments indicate that the actin submembrane cytoskeleton contributes to hindering the translational mobility of both receptor and cholesterol molecules, with only a minority (~10-20%) fraction of the motion-restricted molecules hopping from one compartment to another.

Concluding remarks

The observation of receptor and cholesterol trajectories with the spatiotemporal resolution afforded by MINFLUX opens new possibilities to explore lipid-membrane protein interactions in live cells. The observation that the receptor and cholesterol trajectories overlap spatially and temporally is a novel finding that expands our knowledge of the mutual interactions between a neurotransmitter receptor protein and the neutral lipid at the plasma membrane (Barrantes 2023, 2024b). Achieving sub-millisecond time resolution and spatial resolving power of ~ 7 nm, the MINFLUX study surpasses the capabilities of traditional camera-based techniques like STORM, unveiling individual step characteristics, dissecting the lifetimes of the free walks and confinement sojourns, and several other metrics of the single-molecules trajectories.

Man-tailored depletion/enrichment of cholesterol content confirms the interplay between the neutral lipid and nAChR dynamics, and CK-666 inhibition of Arp2/3 actin nucleation provides evidence of the additional, albeit minor contribution of the submembrane cortical actin meshwork in restricting the translational motion of the nAChR and cholesterol. The directionality (turning angle) analysis of the SMTs revealed distinct preferences for step correlations, especially within confined sojourns, where steps were markedly anticorrelated in the subdiffusive sojourns. The ability to dissect free walks from confined sojourns facilitated the recognition of overlaps of nAChR and cholesterol trajectories in confined and non-confined areas, thus providing one of the strongest pieces of evidence of the tight relationship and mutual interactions between cholesterol and nicotinic receptor dynamics.

Methods

Materials

Fluorescent polyethylene glycol K114-labelled cholesterol (fPEG-Chol, trademarked as AbberiorStar Red-PEG-Chol) was purchased from Abberior GmbH, Göttingen, Germany. A stock solution (1 mg/mL) in

ethanol was kept at -20°C. CF[®]640R-labelled α -bungarotoxin (CF[®]640R-BTX excitation/emission max.: 642/663 nm) and CF[®]680R-labelled α -bungarotoxin (CF[®]680R, excitation/emission max.: 680/701 nm), were purchased from Biotrend GmbH, Köln, Germany. Methyl- β -cyclodextrin (product No. 332615), cholesterol-methyl- β -cyclodextrin complex (water-soluble cholesterol, product No. C4951) and the Arp2/3 inhibitor I CK-666 (product No. 182 515) were purchased from Sigma-Aldrich. Colloidal gold nanoparticle fiducials (150 nm) were purchased from BBI Solutions, Crumlin, U.K.

Sample preparation

CHO-K1/A5 cells were grown as originally reported (Roccamo et al. 1999). Following a washing step with cell culture medium without phenol red, cells were incubated for 5 min on ice with a blocking medium consisting of 1 mg/mL bovine serum albumin in PBS, followed by 30 min labelling on ice with either CF[®]640R-BTX, CF[®]680R-BTX, fPEG-Chol or a combination of CF[®]680R-BTX + fPEG-Chol in 1 mL of culture medium. After several initial assays the final concentration of the two fluorescent α -bungarotoxin derivatives was kept at 500 pM, and that of the fluorescent cholesterol analogue at 1 nM. Gold fiducials were added to the sample chamber containing the coverslip-adhered cells and allowed to settle while the sample was mounted in the microscope stage.

CDx treatment was performed as previously reported (Borroni et al. 2007) by incubation of the cells with 10 mM CDx in PBS for 20 min on ice before the incubation with the fluorescent dyes. Cells were treated with 10 mM CK-666 in PBS for 15 min at room temperature. Soluble cholesterol was added at two concentrations (50 and 100 nM).

MINFLUX experiments

Measurements were performed on a MINFLUX setup from Abberior Instruments GmbH (Göttingen, Germany), based on the iterative localization approach (Schmidt et al. 2021). To localize the emitter and

collect emitted photons we used a hexagonal-shaped beam pattern of diameter L (see Table 3) with points equidistant to the beam centre, which features zero intensity. The use of a circularly-polarized doughnut-shaped zero-intensity beam pattern provided an orientation-independent localization of the fluorophore emission dipoles. The position of the emitter was refined through an iterative sequence of consecutive pattern scanning operations that zoomed-in onto the emitter with decreasing pattern diameter (L) values, as shown in Table 3. These sequences are performed automatically by the microscope control electronics. In short, MINFLUX repeats the first iteration until a fluorescence signal is detected, aligns the zero-intensity centre of the excitation beam, which is precisely known, with the position of the emitter, at which point the microscope will seek to improve the localization by performing successive iterations (4 in the present work) with an increasingly small beam diameter, thus bringing the doughnut beam zero centre closer to the emitter at each iteration. Tracking is then achieved by repeating the last iteration until the signal is lost, at which point the process starts anew. Other essential parameters for the 2D scanning sequence, provided by the manufacturer, are listed in Table 3 below. Another variable that was modified between pattern iterations is the excitation laser power. This was increased in discrete steps between iterations with the laser power multiplier parameter listed in the table, which refers to a reference excitation power of $1.78 \mu\text{W}$ at the exit of the objective. The sampling rate of the tracking measurements performed with this method was between 5 and 10 kHz. The localization precision of the experiments was calculated from the non-linear fitting of the MSD curves (explained in two sections further on). We obtained $\sigma = 7 \pm 1 \text{ nm}$. All ROIs were manually identified using the MATLAB roipoly function implemented in Python:

<https://github.com/jdoepfert/roipoly.py>.

Table 3. MINFLUX scanning parameters for 2D tracking of mobile molecules

2D tracking	1 st iteration	2 nd iteration	3 rd iteration	4 th iteration
Width of the hexagon (nm)*	284	302	150	100
Pattern shape	hexagon	hexagon	hexagon	hexagon

Collected photons (counts)	40	20	20	20
Laser power multiplier (fold)	1	1	2	3
Pattern dwell time (μs)	100	100	100	100
Pattern repeat (times)	1	1	1	1
CFR	-1.0	-1.0	0.8	-1
Background threshold (kHz)	15	30	30	50

*Sometimes referred to as “L”, the diameter of the interrogating MINFLUX beam is here called “width of the hexagon” to avoid confusion.

Initial analysis of MINFLUX experimental series and immobile molecule exclusion

Tracks with irregular time intervals were analysed using object- and array-oriented routines written in Python. A few tracks consisting of only one step (“frustrated” events) were detected and excluded from further analysis. Each complete dataset, consisting of several thousand steps, was analysed including all validated trajectories; the same set of data was next subjected to the criterion of (Golan and Sherman 2017) to exclude immobile particles from the single-molecule tracking (SMT) analysis as in (Mosqueira et al. 2018, 2020). The above criterion is based on the ratios of the radius of gyration R_g^2 , defined as:

$$R_g^2 = \frac{1}{N} \sum_{i=1}^N (\vec{r}_i - \langle \vec{r} \rangle)^2 \quad (\text{Eq. 1})$$

where $\vec{r}_i$ is the coordinate at time step i out of N total time steps and the mean step size $\langle |\Delta r| \rangle$ of the recorded trajectories. Trajectories classified as immobile using the aforementioned criterion were not analyzed further.

In the case of the fPEG-Chol alone and fPEG-Chol(+CF®640R-BTX), the complete raw datasets were excluded from threshold selection, because the distribution of the normalized ratios in these datasets is apparently uniform. Including these ratios into the threshold selection (which depends on the 95% confidence intervals of the mean) would select an overestimated value leading to incorrectly classifying trajectories as immobile. Suppl. Figure 2 shows the histogram of the normalized ratios and examples of trajectories classified into immobile and mobile.

Discrimination between fPEG-Chol and CF[®]680R-BTX in double-labelling experiments

In the case of samples co-labelled with both fPEG-Chol and CF[®]680R-BTX, to discriminate between the two signals we used a Detector Channel Ratio (DCR) criterion, as schematically shown in Figure 7 below. The DCR criterion was applied to all individual points of the trajectories. If the average DCR of a trajectory was above a certain threshold, the particle was assigned to fPEG-Chol tracks. Otherwise, the track was assumed to correspond to CF[®]680R-BTX. To establish the threshold, we plotted histograms of the DCR when both probes were recorded separately, as shown in Suppl. Figure 1. A difference between the two mean distributions was observed, and a threshold ~ 0.55 was applied as it discriminated well between the two distributions.

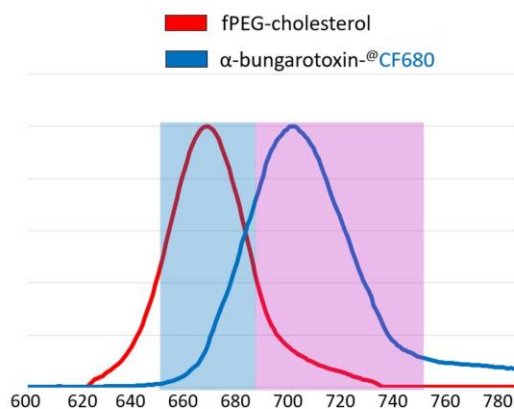


Figure 7. Schematic depiction of the detector channel ratio (DCR) discrimination in co-labelling experiments. fPEG-Chol had an excitation maximum of 638 nm and an emission maximum of 655 nm; CF[®]680R-BTX maxima were at 680 nm and 701 nm. The DCR is defined as $I_1/(I_1+I_2)$, where I_1 and I_2 are the intensities recorded in two wavelength-distinct detection channels (1, fPEG-cholesterol channel: 650-685 nm; 2, CF[®]680R-BTX channel: 685-750 nm), and represented as the effective counts at offset (ECO, referring to the photon counts collected at the outer points of the MINFLUX scanning pattern corrected for background contributions).

Turning Angle Analysis

Turning angle is conventionally measured for increasing time lags, Δ (Sadegh et al. 2017; Burov et al. 2013). As the intervals in the MINFLUX series are not regular, step lags were used instead of time lags. Hence, the relative angle $\theta(i; \Delta)$ distended between successive steps is defined as:

$$\cos[\theta(i; \Delta)] = \frac{\mathbf{V}(i; \Delta) \cdot \mathbf{V}(i + \Delta; \Delta)}{|\mathbf{V}(i; \Delta)| |\mathbf{V}(i + \Delta; \Delta)|} \quad (\text{Eq. 2})$$

such that $\mathbf{V}(i; \Delta) = \overrightarrow{x'_{i+\Delta}} - \overrightarrow{x'_i}$. Basically, $\theta(i; \Delta)$ is the angle between $\mathbf{V}(i; \Delta)$ and $\mathbf{V}(i + \Delta; \Delta)$. The formula was iterated through all trajectory positions whenever possible. The probability density function (PDF) for increasing step lags was obtained from the histogram with a bin width = 10. The algorithm described in (Burov et al. 2013) was vectorized to improve computing times.

Analysis of trajectory kinetic parameters

For a given trajectory j and number of $N_{m\Delta_t}$ displacements intervals between $\Delta_0 + (m - 1)\Delta_t$ and $\Delta_0 + m\Delta_t$, the Time-Averaged MSD (TA-MSD) is defined as:

$$TA - MSD(t_{lag} = m\Delta_t) = \langle \Delta r^2(t_{lag} = m\Delta_t) \rangle_T = \frac{1}{N_{m\Delta_t}} \sum_{i=1}^{N_{m\Delta_t}} d_j(t_i, t_i')^2 \quad (\text{Eq. 3})$$

$$d_j(t_1, t_2) = [x_j(t_2) - x_j(t_1)]^2 + [y_j(t_2) - y_j(t_1)]^2 \quad (\text{Eq. 4})$$

where x_j and y_j are the coordinates of the trajectory on the 'x' and 'y' axes, $\Delta_0 + (m - 1)\Delta_t < t_i' - t_i < \Delta_0 + m\Delta_t$, $\Delta_t = 132 \mu s$, $\Delta_0 = 84 \mu s$ and i the interval number. To calculate the TA-MSD we modified the Python implementation of ref. (Reina et al. 2021) to be suitable for irregular intervals. The only difference of this expression with the one presented in (Rickert et al. 2024) is that displacements are binned into intervals of width Δ_t starting in Δ_0 due to the variable sampling rate in MINFLUX setup of this work. The TA-MSD was fitted to the function:

$$\langle \Delta r^2(t_{lag} = m\Delta_t) \rangle_T = \Gamma * (m\Delta_t)^{\beta-1} * 2 * n_d * \Delta_t * m * \left(1 - \frac{2R}{m}\right) + 2 * n_d * \sigma^2 \quad (\text{Eq. 5})$$

where R is a factor that accounts for the motion blur caused by molecular motion during the time of the measurement, σ is the dynamic localization uncertainty, and n_d is the number of dimensions in which

the diffusion takes place. Here, for molecules diffusing in the plane of the membrane, $n_d = 2$ for 2D trajectories. In addition, we set $R = 1/6.2$ (Balzarotti et al. 2017) which assumes homogeneous illumination during the experiments (Rickert et al. 2024; Reina et al. 2022).

The anomalous exponent (β), the transport coefficient (Γ) and σ were obtained by fitting MSD points up to 50 ms step points of the individual time-averaged MSDs (TA-MSD). If $R = 0$ and $\Delta = 0$, we found that, compared to the formula $\langle \Delta r^2(t_{lag} = m\Delta_t) \rangle_T = K_\beta t^\beta$ (Mosqueira et al. 2020) we obtained $K_\beta = \Gamma * 2 * n_d$. Trajectories were separated into subdiffusive ($\beta \leq 0.9$), Brownian ($0.9 < \beta \leq 1.1$) and superdiffusive ($\beta > 1.1$). This classification criterion is a simplified version of the one outlined in (Mosqueira et al. 2018, 2020). Finally, all fittings with a residual standard deviation greater than 1000 nm² or MSD curves with $\leq 50\%$ of available points were rejected.

Tracking recurrence analysis

Trajectories were found to be interrupted by periods of confinement (i.e., a nAChR particle repeatedly “visited” the same area). Quantitative metrics of the confinement sojourns were obtained using the algorithm of Krapf and coworkers (Sikora et al. 2017). Such algorithm “draws” a circular area between consecutive localizations and measures the number of times that the particle calls on the given area. If the particle visited the area more than V_{th} times, the particle was considered transiently confined. Initially, this threshold, which depends on the experimental conditions, was set to a value of 11, as used for experimental data in ref. (Sikora et al. 2017). This threshold erroneously detected confinement in places where, through visual inspection, the moving particle appeared to follow a Brownian behaviour. V_{th} was therefore increased to a value = 33. To prevent false state transitions within a trajectory, we took the sequence of states assigned by the algorithm and divided it into non-overlapping windows of 3 steps. Next, each step within the window was reassigned to the state that occurred most frequently within that specific window. To analyse confinement sojourn parameters, trajectories were segmented

into sub-trajectories between confined and mobile states. The shape of the confinement area was fitted using an algorithm that determines the eccentricity and length of the semi-axes defined by an ellipse fitted on these data points. The area of confinement zones is equal to the area defined by their convex hull.

Spatial overlap in confined and non-confined regions of the trajectories

In cells co-labelled with fPEG-Chol and the nAChR fluorescent antagonist CF[®]680R-BTX, the trajectories of the two probes frequently intersected. To determine whether these intersections occurred in confined or non-confined portions of the trajectories and how recurrent this phenomenon was: a) the confinement areas of each CF[®]680R-BTX trajectory ($C_{CF^{®}680R-BTX}$) were considered to overlap with the confinement areas of fPEG-Chol trajectories ($C_{CF^{®}680R-BTX} \cap fPEG-Chol$) if their convex hulls intersected. An apparent overlap coefficient, $C = C_{CF^{®}680R-BTX} \cap fPEG-Chol / C_{CF^{®}680R-BTX}$ was also calculated for each trajectory. This metric can also be defined as the ratio $C = C_{fPEG-Chol} \cap CF^{®}680R-BTX / C_{fPEG-Chol}$; b) to characterize the overlap in non-confined, free-walk portions of the trajectories, the spatial proximity between the individual steps of fPEG-Chol and CF[®]680R-BTX was estimated using k-d tree structures (Bentley 1975). A k-d tree is a multidimensional tree-like structure that enables efficient spatial queries, such as the proximity between trees. The k-d tree implementation included in the SciPy software package (Virtanen et al. 2020) and the polygon intersection algorithm from Shapely (Gillies 2023) were used to this end. Steps from two distinct k-d trees are considered close if they are within a distance $< r$. This parameter was estimated for each CF[®]680R-BTX and fPEG-Chol non-confined portions by querying their corresponding k-d trees using a value of $r = 10$ nm. Steps were considered to overlap if they were within a distance smaller than this value, equivalent to the localization precision.

Spatial confinement overlap significance test

Measuring confinement overlaps using the convex hull was found to be too slow for estimating the statistical significance of the overlap between confinement portions. Instead, we resorted to the Intersection over Union (IoU) metric in the following manner. First, the ROI was divided into square cells of 7 x 7 nm. The ROIs were then converted into two matrices: one for fPEG-Chol and another for CF[®]680R-BTX. The fPEG-Chol matrix was assigned a value of 1 if there was at least one fPEG-Chol confined localization in that cell. The IoU was defined as the intersection (cells with both fPEG-Chol and CF[®]680R-BTX) divided by the union (cells with either fPEG-Chol or CF[®]680R-BTX) of the two matrices. To determine if the IoU was significant, we used the Andi-dataset package (Muñoz-Gil 2023) to simulate trajectories within a ROI of the same dimensions. First, we defined the dimensions of the ROIs where the trajectories are located. Next, for each actual experimental trajectory a simulated trajectory was placed within the ROI, ensuring it moved through confinement zones randomly placed within the ROI. These zones had the same radius as those obtained previously using the recurrent algorithm, covering ~10% of the ROI area. This process was repeated for CF[®]680R-BTX trajectories. The IoU of the simulated experiment was measured next. The resulting calculated IoUs were then compared with the experimental IoU values using the percentile $\alpha' = 1 - \alpha$ percentile (significance level of $\alpha = 0.05$) of the randomly simulated IoUs. If the actual IoU was larger than the α' percentile, the overlap was significantly higher than that expected by chance. The same procedure was applied to non-confined portions. This analysis ensured that the significance of the overlap was assessed rigorously through statistical testing, providing robust conclusions about the spatial relationships between the two probe channels in the ROI.

Particle dynamics characterization between free and hop diffusion

To categorize the diffusional behaviour between free and hop diffusion, we followed the workflow described in (Rickert et al. 2024), which is based on (Reina et al. 2022). Trajectories were analysed by fitting their MSD curves to two different models: MSD_{free} and MSD_{hop} . Prior to the analysis proper, all

trajectories were truncated to segments of 500 localizations each to ensure uniform statistical sampling for each trajectory.

If the particles diffuse freely in 2D, the MSD is described as:

$$\text{MSD}_{\text{free}}(n\Delta_t) = 2n_d D n\Delta_t \quad (\text{Eq. 6})$$

MSD_{hop} combines MSD_{free} and MSD_{conf} (i.e., the MSD describing a spatially confined particle). MSD_{hop} is described as:

$$\text{MSD}_{\text{hop}} = \text{MSD}_{\text{free}}(n\Delta_t) + \frac{D_\mu - D_M}{D_\mu} \text{MSD}_{\text{conf}}(n\Delta_t) \quad (\text{Eq. 7})$$

where D_μ and D_M are the short-term (*intra-compartmental*) and long-term (*inter-compartmental*) diffusion coefficients, respectively (Rickert, 2024). Analytically speaking, if the molecule diffuses freely, $D_M = D_\mu$ ($\frac{D_\mu - D_M}{D_\mu} = 0$); if the molecule is totally confined within a single confinement region along its lifetime, $D_M = 0$ (Reina et al. 2022).

The type of diffusion described by MSD_{hop} occurs when confinement regions are partially permeable, and the molecules have a probability $p_{\text{hop}} > 0$ to jump to another confinement region. Following the approaches of (Daumas et al. 2003) and (Rickert et al. 2024), MSD_{conf} can be formulated as:

$$\text{MSD}_{\text{conf}}(n\Delta_t) = A \left[1 - \exp\left(-\frac{n\Delta_t}{\tau}\right) \right] \quad (\text{Eq. 8})$$

where A is the average distance between two randomly chosen points within the confinement regions and τ is the equilibration time at which the effect of boundaries appears and the MSD reaches its plateau region (Manzo and Garcia-Parajo 2015). Since Hell and coworkers defined the expressions for 3D trajectories in a 3D space and hence the confinement regions have a volume (Rickert et al. 2024), the expressions had to be adapted to two dimensions. Confined diffusion can be restricted to a square domain of side L , and hence $A = L^2/3$ where L is the apparent confinement domain size (Destainville and Salome 2006). The equilibration time, $\tau = \frac{A}{4D_\mu}$, such that D_μ is the diffusion coefficient that dominates the diffusion within the confinement region (Daumas et al. 2003) results in:

$$\tau = \frac{L^2}{12 D_\mu} \quad (\text{Eq. 9})$$

A squared confined domain L is taken for the sake of simplicity, but any 2D geometric shape can be considered (Destainville and Salome 2006) in the MSD_{conf} :

$$\text{MSD}_{\text{conf}}(n\Delta_t) = \frac{L^2}{3} \left[1 - \exp\left(-\frac{12 D_\mu n\Delta_t}{L^2}\right) \right] \quad (\text{Eq. 10})$$

Having obtained the expressions for MSD_{conf} and MSD_{free} , MSD_{hop} can be rewritten for 2D tracks as:

$$\begin{aligned} \text{MSD}_{\text{hop}}(n\Delta_t) &= \text{MSD}_{\text{free}}(n\Delta_t) + \frac{D_\mu - D_M}{D_\mu} \text{MSD}_{\text{conf}}(n\Delta_t) = \\ &= 2n_d D_M n\Delta_t + \frac{D_\mu - D_M}{D_\mu} \frac{L_{\text{hop}}^2}{3} \left[1 - \exp\left(-\frac{12 D_\mu n\Delta_t}{L_{\text{hop}}^2}\right) \right] = \\ &= 2n_d n\Delta_t \left\{ D_M + \frac{D_\mu - D_M}{D_\mu} \frac{L_{\text{hop}}^2}{6 n_d n\Delta_t} \left[1 - \exp\left(-\frac{12 D_\mu n\Delta_t}{L_{\text{hop}}^2}\right) \right] \right\} \quad (\text{Eq. 11}) \end{aligned}$$

such that diffusion coefficients D_μ and D_M are separated by a length scale L_{hop} (Rickert et al. 2024).

Application of this analysis returned in some cases a $D_M \cong 0$. For this reason, we decided to include MSD_{conf} within the scope of the analysis.

Thus far MSD_{hop} and MSD_{free} do not include corrections for measurement artifacts. To account for these, MSD_{hop} and MSD_{free} were rewritten (Rickert et al. 2024) as:

$$\text{MSD}_{\text{free}}(n\Delta_t) = 2n_d D \Delta_t (n - 2R) + 2n_d \sigma^2 \quad (\text{Eq. 12})$$

$$\text{MSD}_{\text{hop}} = 2n_d \Delta_t \left\{ D_M + \frac{D_\mu - D_M}{D_\mu} \frac{L_{\text{hop}}^2}{6 n_d n\Delta_t} \left[1 - \exp\left(-\frac{12 D_\mu n\Delta_t}{L_{\text{hop}}^2}\right) \right] \right\} (n - 2R) + 2n_d \sigma^2 \quad (\text{Eq. 13})$$

Because MSD_{conf} is equal to MSD_{hop} when $D_M = 0$, MSD_{conf} is rewritten as:

$$\text{MSD}_{\text{conf}} = \frac{L^2}{3} \left[1 - \exp\left(-\frac{12 D_\mu n\Delta_t}{L^2}\right) \right] (n - 2R) + 2n_d \sigma^2 \quad (\text{Eq. 14})$$

In the case of MSD_{free} , the free parameters are D and σ . In the case of MSD_{hop} , the free parameters are D_M , D_μ , L_{hop} , and σ . Finally, in the case of MSD_{hop} , the free parameters are D_μ , L , and σ . MSD_{free} derived in Eq. 12 has a different nomenclature but is essentially the same expression employed by Hell and coworkers (Balzarotti et al. 2017).

To classify trajectories, the following steps were followed: (i) an MSD curve was calculated for each trajectory; (ii) the initial 20% data points of the MSD curves were fitted to MSD_{free} , MSD_{hop} , and MSD_{conf} solving three non-linear optimization problems; the free parameters were adjusted such that the sum of the squared residuals (SSR) of both regressions on the observed data were minimized. We used the python-based software package SciPy to solve this optimization problem 100 times (the solution with the least SSR was picked for each case). MSD points were sampled logarithmically to emphasize the fitting to the initial 20% MSD points. (iii) Next, the Bayesian Information Criterion (BIC) was calculated for both fits:

$$\text{BIC} = n * \ln \left(\frac{\text{SSR}}{2} \right) + k * \ln (n) \quad (\text{Eq. 15})$$

where n and k are the number of data points and number of fit parameters, respectively.

In Rickert et al. (Rickert et al. 2024) the MSD curves are fitted to MSD_{hop} with constraint $D_{\mu} > 5D_M$ to find substantial differences between the two diffusion coefficients. However, this constraint artifactually imposes a minimum value of 5 for $S_{\text{conf}} = D_{\mu}/D_M$, which is defined as the confinement strength (i.e., the normalized residence time of a confined molecule) and $S_{\text{conf}} > 1$ (Wieser, 2015) and hence this constraint was removed, leaving $D_{\mu} > D_M$. Table 4 depicts the boundaries of the initial free parameter values for each MSD model to solve the optimization problem.

Table 4. Boundaries for initializing free parameters in the optimization process.

Optimization parameters		
Parameters	Lower Bound	Upper Bound
D (Eq. 12)	100 nm ² s ⁻¹	100,000 nm ² s ⁻¹
σ (Eq. 12, 13, 14)	10 nm	100 nm
D_M (Eq. 13)	100 nm ² s ⁻¹	100,000 nm ² s ⁻¹
D_{μ} (Eq. 13)	100 nm ² s ⁻¹	100,000 nm ² s ⁻¹
L_{hop} (Eq. 13)	10 nm	1000 nm
$D_{\mu,\text{conf}}$ (Eq. 14)	100 nm ² s ⁻¹	100,000 nm ² s ⁻¹
L (Eq. 14)	10 nm	1000 nm

Statistical Analysis

All graphs and statistics were prepared following the guidelines of ref. (Lord et al. 2020). All reported mean values and statistical tests were conducted on ROI averages. To compare distributions, we used the Kolmogorov-Smirnov (KS) test for two samples (null hypothesis was rejected for p-value <0.05). To compare three or more distributions, the Kruskal-Wallis test was used. Statistical analysis was carried out using GraphPad Prism 8.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Code availability

Python code for the analysis of the data is available at the public repository

https://github.com/lucasSaavedra123/minflux_analysis and deposited at

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Author contributions

F.J.B. and C.E. conceived the project; F.R. and F.J.B. performed the live-cell experiments; L.A.S. wrote the analysis code. L.A.S. and F.J.B. analysed the data; F.J.B wrote the manuscript with feedback from all the authors; F.J.B. edited the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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